NOAA FISHERIES

Regulatory Impact Review for a Rule to Implement Decisions of the Western and Central Pacific Fisheries Commission for:

Bigeye Tuna Catch Limits in Longline Fisheries for 2015

RIN 0648-BF19

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1. INTRODUCTION

This document is a regulatory impact review (RIR) prepared under Executive Order 12866, "Regulatory Planning and Review." Executive Order 12866 requires that the economic impacts of proposed government regulations on the national economy be assessed before implementation. EO 12866 states that "[i]n deciding whether and how to regulate, agencies should assess all costs and benefits of available regulatory measures, including the alternative of not regulating" (EO 12866, Section 1). The emphasis of the analysis is on expected changes in net benefits that occur as a result of the proposed management measures. The government should choose only those sets of regulations that produce positive benefits while considering social and distributional effects (but certain statutes and international obligations sometimes compel the government to take specific actions despite expected economic and social outcomes). NMFS requires that this analysis be done through a regulatory impact review (RIR) for all regulatory actions that are of public interest. The RIR also includes analysis of distributive impacts and the costs of government administration and private compliance with the proposed measures.

This RIR is for a NMFS final rule (to be issued without a proposed rule) that would establish a limit of 3,502 metric tons (mt) of bigeye tuna that may be captured in the area of application of the Convention on the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (Convention Area) by longline gear and retained on board by fishing vessels operating as part of a fishery of the United States. The action would be taken to implement provisions of a conservation and management measure (CMM 2014-01) adopted in December 2014 by the Commission for the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (Commission or WCPFC), as described in section 2.

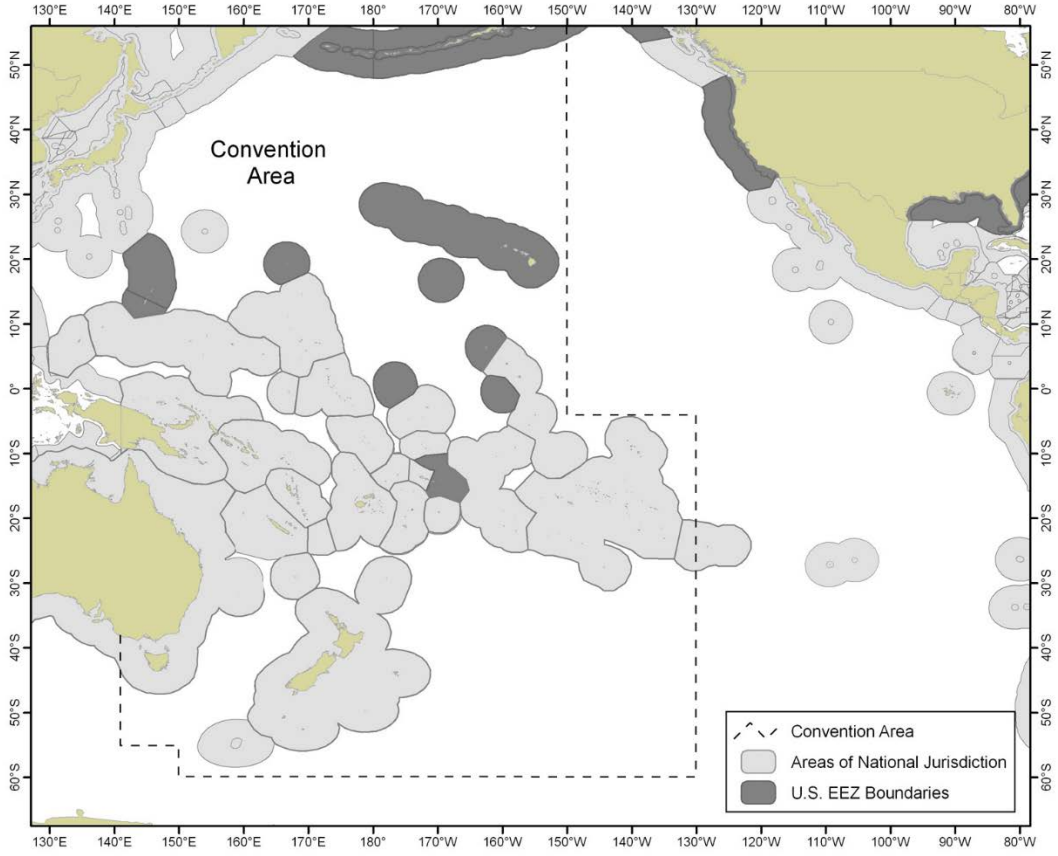
NMFS' final rule would be issued under authority of the Western and Central Pacific Fisheries Convention Implementation Act (16 U.S.C. § 6901 *et seq*.) (WCPFC Implementation Act). The WCPFC Implementation Act authorizes the Secretary of Commerce (Secretary) to promulgate regulations that are needed to carry out the international obligations of the United States under the Convention and the Act, including the decisions of the WCPFC. The Secretary is directed to consult with the Secretary of State and the agency in which the U.S. Coast Guard is operating in promulgating regulations. The authority to promulgate regulations has been delegated to NMFS.

The WCPFC was established under the Convention on the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (Convention). The WCPFC is comprised of the Contracting Parties to the Convention on the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (Convention) and fishing entities that have agreed to be bound by the regime established by the Convention (formally, Commission "Members"). Other entities that participate in the WCPFC include Participating Territories and Cooperating Non-Members. Cooperating Non-Members are admitted on a yearly basis. All these WCPFC participants are generally subject to the obligations set out in conservation and management measures, and they are collectively referred to as WCPFC "members" in this RIR.

The United States ratified the Convention on the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (Convention) in 2007.[1] The area of application of the Convention (Convention Area), which encompasses the WCPO, is shown in Figure 1.

The Convention text indicates that the agreement is focused on highly migratory fish species (HMS) and stocks thereof within the Convention Area (see the Convention text for the specific HMS covered).[2] The Convention provides for the conservation and management of target stocks, non-target species, and species belonging to the same ecosystem or dependent upon or associated with the target stocks.

Figure 1. The Convention Area - high seas (in white); U.S. Exclusive Economic Zone (EEZ) (in dark gray); and foreign jurisdictions ("claimed maritime jurisdictions," in light gray).



Source: NMFS Pacific Islands Regional Office.

[1] The Convention was opened for signature in Honolulu on September 5, 2000, and entered into force in June 2004; the Convention entered into force for the United States in 2007. The full text of the Convention is available at: http://www.wcpfc.int/key-documents/convention-text.

[2] Though not specifically stated in the Convention text, it has also been agreed that southern bluefin tuna (*Thunnus maccoyii*) in the Convention Area will continue to be solely managed by the Commission for the Conservation of Southern Bluefin Tuna.

2. OBJECTIVES

The objective of this action is to satisfy the international obligations of the United States, as a Contracting Party to the Convention, with respect to specific decisions of the Commission, as described below.

At its Eleventh Regular Session, in December 2014, the Commission revised its CMM, "Conservation and Management Measure for Bigeye, Yellowfin and Skipjack Tuna in the Western and Central Pacific Ocean." The latest CMM, CMM 2014-01, is generally applicable for the 2015-2017 calendar year period, and only contains minor modifications to CMM 2013-01, the predecessor CMM for the three main tropical tunas (skipjack tuna (*Katsuwonus pelamis)*, yellowfin tuna (*Thunnus albacares)*, and bigeye tuna (*Thunnus obesus*)). CMM 2014-01 includes provisions for purse seine vessels and longline vessels. The CMM's provisions for purse seine vessels include limits on the allowable level of fishing effort, restrictions on the use of fish aggregating devices (FADs), requirements to retain all catches of skipjack tuna, yellowfin tuna, and bigeye tuna, with some exceptions, and a general provision not to increase catches of yellowfin tuna. The CMM's provisions for longline vessels include catch limits for bigeye tuna and a general provision not to increase catches of yellowfin tuna. CMM 2014-01's stated general objective is to ensure that the stocks of bigeye tuna (*Thunnus obesus*), yellowfin tuna (*Thunnus albacares*), and skipjack tuna (*Katsuwonus pelamis*) in the WCPO are, at a minimum, maintained at levels capable of producing their maximum sustainable yield as qualified by relevant environmental and economic factors. The CMM includes specific objectives for each of the three stocks: for each, the fishing mortality rate is to be reduced to or maintained at levels no greater than the fishing mortality rate associated with maximum sustainable yield.

The provisions of CMM 2014-01 that are being implemented in the final rule considered here are limits on catches of bigeye tuna in longline fisheries for 2015. Under CMM 2014-01 the applicable longline bigeye tuna catch limit for the United States in 2015 is 3,554 mt. In addition, the CMM states that any catch overage in a given year shall be deducted from the catch limit for the following year. This provision regarding overages was also in CMM 2013-01, the predecessor to CMM 2014-01, so it pertains to the formulation of the catch limit for 2015. NMFS has determined that the 2014 bigeye tuna catch limit was exceeded by 52 mt, so the catch limit for 2015 will be 3,502 mt. No limits are required for the longline fisheries of the U.S. territories that participate in the Commission ("participating territories"), American Samoa, the Commonwealth of the Northern Mariana Islands, and Guam. As in previous rules to implement similar Commission decisions, the main consequence of reaching the longline bigeye tuna catch limit would be a prohibition on the retention of bigeye tuna in the Convention Area for the remainder of the limit period. Under existing regulations at 50 CFR 300.224(e), after NMFS determines that the limit is expected to be reached by a specific future date, NMFS will publish a notice in the *Federal Register* announcing that specific prohibitions regarding the retention of bigeye tuna will be in effect starting on that specific future date. The notice will be published at least seven calendar days in advance of that specific future date.

NMFS anticipates proposing, in a separate rulemaking, a framework under which NMFS would implement various requirements adopted by the Commission, including specifying longline catch limits (RIN 0648-BE84). NMFS anticipates implementing the longline bigeye tuna catch limits

under CMM 2014-01 for years 2016 and beyond pursuant to the framework, but would implement the 2015 limit according to this final rule. The proposed action to establish the framework has not been published at the time of writing this RIR, but is anticipated to consist of a proposed rule that would, among other things, establish a framework under which NMFS would specify limits on fishing effort and catches, as well as spatial and temporal restrictions on particular fishing activities and other requirements, in U.S. fisheries for highly migratory fish species in the western and central Pacific Ocean (WCPO). NMFS would issue the specifications as needed to implement conservation and management measures adopted by the Commission for the Conservation and Management of Highly Migratory Fish Stocks in the Western and Central Pacific Ocean (Commission or WCPFC).

NMFS is establishing the 2015 longline bigeye tuna catch limit in this final rule instead of proposing it in the other regulatory action (RIN 0648-BE84) for timing reasons. The most recent available information on bigeye tuna catches in the U.S. longline fishery so far in 2015 indicates that catches are accumulating substantially faster than in previous years, and that the level of 3,502 mt is likely to be reached in August, before NMFS is able to consider public comments on the proposed rule and specifications (RIN 0648-BE84) and issue a final rule and final specifications. Thus, NMFS intends to issue this final rule before issuing the final rule and specifications for RIN 0648-BE84.

3. DESCRIPTION OF AFFECTED FISHERIES

The final rule would affect the U.S. longline fisheries in the Convention Area, particularly the Hawaii-based longline fisheries. These fisheries are described in a programmatic environmental assessment prepared for implementation of Commission decisions on tropical tuna stocks in the WCPO (NMFS 2015b; "2015 PEA"), including characteristics of the fleets, fishing practices, the current management regime, and recent performance of the fisheries in terms of fishing effort and catches. Chapter 3 of the 2015 PEA is incorporated here by reference. One aspect of the management regime that is particularly pertinent to analysis of the final rule is described in detail below.

In 2011, Public Law 112-55, 125 Stat. 552 *et seq*., the Consolidated and Further Continuing Appropriations Act (CFCAA), 2012, was enacted. Section 113 of the CFCAA allowed the U.S. participating territories to use, assign, allocate, and manage catch or effort limits agreed to by the WCPFC through fishing arrangements with fishing vessels of the United States to support fisheries development in the territories ("Section 113 arrangements"), and directed NMFS to attribute pelagic MUS catches made by such vessels to the U.S. participating territory to which the arrangement applies. In 2013, the Section 113 provisions of the CFCAA were extended through 2013.

The CFCAA also directed the Western Pacific Fishery Management Council to recommend an amendment to the Fishery Ecosystem Plan for Pelagic Fisheries of the Western Pacific Region (Pelagics FEP) that would essentially continue the mechanism established in Section 113 of the CFCAA. Amendment 7 to the Pelagics FEP was approved by the Secretary of Commerce on March 28, 2014, and the regulations implementing the amendment went into effect October 24,

2014 (final rule published October 28, 2014; 79 FR 64097). The framework established under Amendment 7 authorizes the governments of American Samoa, the CNMI, and Guam to allocate a portion of any catch or effort limits specified for them to a U.S. fishing vessel or vessels through a specified fishing agreement ("Amendment 7 agreement"). Regulations at 50 CFR 665.819(c) establish procedures and eligibility requirements for such agreements. The bigeye tuna catch made by a vessel in an Amendment 7 agreement is attributed to the longline fishery of the territory that is party to the agreement, up to a collective limit for each territory that is established on a year-to-year basis. The limit for each territory for 2014 was 1,000 mt. The Western Pacific Fishery Management Council has taken action to recommend that the limits for 2015 and 2016 be set at the same level, 1,000 mt, but the recommendation has yet to be approved and implemented by NMFS.

The specific Section 113 arrangements and Amendment 7 agreements that were in place during 2009-2014 are described in section 6.1.

4. PROBLEM STATEMENT

This RIR considers a final rule that would implement, for U.S. fisheries, specific decisions of the Commission, as described in section 2. The purpose of the final rule is to implement such decisions in a timely and practical manner, in order to contribute to the underlying objectives of the Commission's decisions, as described in section 2. The need for the rule is to satisfy the international obligations of the United States as a Contracting Party to the Convention, pursuant to the authority of the WCPFC Implementation Act.

Prior Rules

NMFS has issued several rules that established longline bigeye tuna catch limits similar to those that would be established under this action. The rules resulted in limits of 3,763 mt in each of the calendar years 2009-2014. The limits have been codified at 50 CFR 300.224. For reference, the rules that established those limits are listed below.

- 2009 final rule to implement, for 2009-2011, the longline bigeye tuna catch limit provisions of CMM 2008-01 (December 7, 2009, at 74 FR 63999; "2009 LL rule").

- 2012 interim final rule to implement, for 2012, the longline bigeye tuna catch limit provisions of CMM 2011-01 (77 FR 51709; August 27, 2012; "2012 LL rule").

- 2013 final rule to implement, for 2013-2014, longline bigeye tuna catch limits in accordance with CMM 2012-01 (78 FR 58240; September 23, 2013; "2013 LL rule").

RIRs were prepared for each of the rules listed above. The action alternatives considered in each RIR are described in section 5.3.

5. ALTERNATIVES

5.1. Alternative 1 (no action)

Under the no-action alternative, no regulatory changes would be made. The existing management regime for the U.S. longline fisheries operating in the Convention Area would remain in effect.

5.2. Alternative 2 (final rule)

The final rule would establish a limit for calendar year 2015 of 3,502 mt of bigeye tuna that may be captured in the Convention Area by longline gear and retained on board by fishing vessels operating as part of a fishery of the United States. Bigeye tuna caught in the longline fisheries of American Samoa, Guam, and the Commonwealth of the Northern Mariana Islands (CNMI) would not be counted against the limit, and vessels participating in those fisheries would not be subject to the restrictions triggered by the limit being reached. The territorial longline fisheries would be distinguished from those of the United States based on a combination of three factors: (1) where the bigeye tuna are landed; (2) the types of Federal longline fishing permits registered to the fishing vessel; and (3) whether the fishing vessel is included in an Amendment 7 agreement. Specifically, all fishing activities by U.S. longline fishing vessels would be considered to be part of a fishery of the United States except as follows (see 50 CFR 300.224(b)-(d)):

(1) Except as provided under paragraphs (2) and (3) below, if catch is landed in American Samoa, Guam, or the CNMI, the catch and associated fishing effort is considered part of a fishery of the territory in which it is landed, provided that: (a) it was not caught using purse seine gear; (b) it was not caught in any portion of the U.S. exclusive economic zone (EEZ) other than the portion of the U.S. EEZ surrounding the territory in which it was landed; and (c) it was landed by a fishing vessel operated in compliance with a valid permit issued under §660.707 or §665.801 of this title.

(2) Except as provided under paragraph (3) below, if catch is made by longline gear by a vessel registered for use under a valid American Samoa Longline Limited Access Permit issued under §665.801(c) of this title, the catch and associated fishing effort is considered part of a fishery of American Samoa, provided that: (a) it was not caught in any portion of the U.S. EEZ other than the portion of the U.S. EEZ surrounding American Samoa; and (b) it was landed by a fishing vessel operated in compliance with a valid permit issued under §660.707 or §665.801 of this title.

(3) If catch or fishing effort is made by a vessel that is included in a specified fishing agreement under §665.819(c) of this title (an Amendment 7 agreement), the catch and associated fishing effort is considered part of a fishery of American Samoa, Guam, or the CNMI, according to the terms of the agreement to the extent the agreement is consistent with §665.819(c) of this title and other applicable laws, provided that: (a) the start date specified in §665.819(c)(9)(i) of this title has occurred or passed; and (b) NMFS has not made a determination under §665.819(c)(9)(iii) of this title that the

catch or fishing effort exceeds any limit allocated to the territory that is a party to the agreement.[3]

Once NMFS determines that the limit is expected to be reached and issues a *Federal Register* notice to that effect under 50 CFR 300.224(e), owners and operators of U.S. longline vessels would be subject to the prohibitions listed below from the closure date through December 31, 2015 (see 50 CFR 300.224(f)). The closure date would be no earlier than 7 days after the closure notice in the *Federal Register*.

(1) Except for fishing vessels operating as part of a territory's longline fishery (see criteria above), a fishing vessel of the United States may not be used to retain on board, transship, or land bigeye tuna captured by longline gear in the Convention Area, except that any bigeye tuna already on board a fishing vessel upon the effective date of the prohibitions may be retained on board, transshipped, and/or landed, to the extent authorized by applicable laws and regulations, provided that they are landed within 14 days after the prohibitions become effective. The 14-day landing requirement does not apply to a vessel that has declared to NMFS, pursuant to 50 CFR 665.803(a) of this title, that the current trip type is shallow-setting.

(2) Bigeye tuna caught by longline gear in the Convention Area may not be transshipped to a fishing vessel unless that fishing vessel is operated in compliance with a valid permit issued under 50 CFR 660.707 or 665.801.

(3) Except for fishing vessels operating as part of a territory's longline fishery (see criteria above), a fishing vessel of the United States may not be used to fish in the Pacific Ocean using longline gear both inside and outside the Convention Area during the same fishing trip, with the exception of a fishing trip during which the prohibitions were put into effect as announced under 50 CFR 300.224(e), in which case the bigeye tuna on board the vessel may be retained on board, transshipped, and/or landed, to the extent authorized by applicable laws and regulations, provided that they are landed within 14 days after the prohibitions become effective. This prohibition does not apply to a vessel for which a declaration has been made to NMFS, pursuant to 50 CFR 665.803(a), that the current trip type is shallow-setting.

(4) Except for fishing vessels operating as part of a territory's longline fishery (see criteria above), and vessels for which declarations have been made to NMFS, pursuant to 50 CFR 665.803(a), that the current trip type is shallow-setting, if a fishing vessel of the United States is used to fish in the Pacific Ocean using longline gear outside the Convention Area and the vessel enters the Convention Area at any time during the same fishing trip, the longline gear on the fishing vessel must, while it is in the Convention Area, be stowed in a manner so as not to be readily available for fishing; specifically, the hooks, branch or dropper lines, and floats used to buoy the mainline must be stowed and not available for immediate use, and any power-operated mainline hauler on deck must be covered in such a manner that it is not readily available for use.

[3] The regulations at 50 CFR §665.819(c) implement Amendment 7 to the Fishery Ecosystem Plan for Pelagic Fisheries of the Western Pacific Region.

5.3. Alternatives excluded from detailed analysis

In the RIR for the 2009 LL rule (NMFS 2009), NMFS considered four action alternatives to implement longline bigeye tuna catch limits of 3,763 mt in each of 2009, 2010, and 2011. The alternatives addressed two aspects – which activities would be prohibited in the Convention Area once the limit is reached, and how to distinguish the longline fisheries of the participating territories (which would not be subject to the limits) from the U.S. longline fishery. Regarding the first aspect, the alternatives included prohibiting both deep-setting and shallow-setting, prohibiting deep-setting but allowing shallow-setting provided that no bigeye tuna or yellowfin tuna are retained, and allowing both types of longlining provided that no bigeye tuna are retained (with some exceptions). The last alternative was ultimately implemented, and continued to be implemented for the limits through 2014. Regarding the second aspect, one alternative would attribute catches among the fisheries based (with some provisos) on the place of landing. The other alternative, which was implemented in the 2009 LL rule and for the subsequent limits through 2014, attribution among the fisheries was determined by both place of landing and permit types; specifically, catches of vessels with American Samoa longline permits were, with some provisos, attributed to the longline fishery of American Samoa.

In the RIR for the 2012 LL rule (NMFS 2012), NMFS considered five action alternatives to implement a longline bigeye tuna catch limit of 3,763 mt for 2012. Four were identical to the four considered for the 2009 LL rule. The fifth, which was implemented, took into account Section 113 of the CFCAA, such that attribution among the fisheries was determined by place of landing, permit types, and whether a vessel was in a Section 113 arrangement.

In the RIR for the 2013 LL rule (NMFS 2013), which established longline bigeye tuna catch limits of 3,763 mt in each of 2013 and 2014, NMFS considered three options regarding when NMFS would start attributing catches to territories under Section 113 arrangements. Those options are not relevant for the action considered in this RIR because the timing of attribution under Amendment 7 agreements is specified in Amendment 7 and its implementing regulations.

Some of the alternatives and variants considered in the RIRs for the previous rules could conceivably be implemented as part of the action considered here. NMFS does not believe that any of those alternatives warrant in-depth analysis again, but the previous RIRs identified above are incorporated here by reference.

6. ANALYSIS OF ALTERNATIVES

Four types of economic effects are analyzed: changes in net benefits, distributional changes in net benefits, changes in income and employment, and cumulative effects.

6.1. Changes in net benefits

Analytical approach:

The analysis is limited to examining changes in net benefits to the United States; changes in net benefits that occur to foreign interests are not relevant in the context of this RIR. Changes in benefits and costs in both the private and public sectors are important with respect to net benefits; effects in both sectors are accounted for in this analysis to the extent possible. In the private sector, benefits may accrue as surpluses to consumers and producers. In the case of fish harvesting operations, producer surplus is reflected in the difference between gross revenues and operating costs. Expected changes in benefits and costs as a result of this action cannot be entirely quantified, so the analysis is mostly qualitative.

The action would be effective only in 2015, so its direct effects on the conduct of fishing vessels would be largely limited to that period and its economic impacts would be accordingly short-lived (but see section 6.4 regarding the cumulative impacts of this action with those of other foreseeable future actions). Thus, there is no need to estimate benefits or costs in future years.

The economic effects of the action could depend greatly on the terms of any Amendment 7 agreements in effect in 2015. Because NMFS cannot predict whether such agreements will be in effect or what all their terms will be, it is difficult to assess the likely impacts of the action. However, it is possible to identify the bounds of the likely range of economic effects. At one extreme of the range, no Amendment 7 agreements would be in place, or their terms would be such that they would not affect the attribution of any bigeye tuna catches. This is referred to in this RIR as the "most-constraint" scenario. At the other extreme, Amendment 7 agreements would be in effect and their terms would be such that the U.S. longline fleet is effectively unconstrained by the catch limits established by this action. In that case, the net economic impacts would be expected to be the same as under the no-action alternative (but there could be some distributional changes in net benefits, described in section 6.2). This is referred to as the "least-constraint" scenario.

Overall benefit of the action:

The action is a conservative one in that it is intended to reduce the fishing mortality rate (i.e., the proportion of a stock that is killed by fishing per unit of time) of a stock (WCPO bigeye tuna) that NMFS has determined to be subject to overfishing (its fishing mortality rate is greater than the rate associated with maximum sustainable yield (MSY). To the extent it does so, it would enhance the stock's productivity, and in turn enhance the ability of the United States to benefit from the stock, be it through producer surplus generated in the U.S. fisheries or consumer surplus generated by both domestically produced and imported tuna from the WCPO.

The main expected direct effect of the action (only under the most-constraint scenario) would be less U.S. longline fishing effort and less bigeye tuna and other species caught by the fishery in the WCPO, and possibly more fishing effort and catch of those species in the eastern Pacific Ocean than would otherwise occur.

Those direct effects can be expected to lead to indirect effects on WCPO stocks of the main target species in the longline fisheries, bigeye tuna, as well as on stocks of fish and non-fish species caught incidentally in the fisheries. Overall, the possible reduction in longline fishing effort could cause a reduction in the fishing mortality rates of these stocks and consequent increases in the stocks' sizes.

In any case, because any effects of the action on fishing mortality rates would be small compared to the stocks' total fishing mortality rates, and because the action would be limited to 2015, any effects on stock sizes would be relatively small and short-lived. As described further in section 6.4, if similar restrictions are implemented in the future, and if the United States and other member nations of the WCPFC impose similar restrictions in their longline fisheries and other fisheries, particularly their purse seine fisheries, there could be greater effects on the sizes of WCPO stocks.

Adult bigeye tuna and yellowfin tuna are considered among the top predators of the tropical or warm pool marine ecosystem (see Chapter 3 of the 2015 PEA for more information). Any effects of the action on the WCPO stocks of these species and other high-tropic-level species could lead to trophic interactive effects. For example, if these stocks are larger than they would be under no-action, there could be increased competition for prey species with other top predators. Larval and juvenile tunas are also sources of food for other marine species, such as fish, seabirds, porpoises, marine mammals, and sharks. Thus, increases in abundance of larval and juvenile tuna could increase the food available for these other species. Because the effects to the tuna stocks are expected to be relatively minor and short-lived, it is unlikely that any substantial or lasting impacts to marine ecosystem would occur as a result of the action.

To gauge the net benefits of the action, the (uncertain and unquantifiable) benefits identified here and in section 6.4 would have to be weighed against the costs of the action. Those impacts are estimated to the extent possible in the sections that follow, specifically in terms of consumer surplus, producer surplus, and public sector costs.

Consumer impacts:

Consumer surplus is the difference between what consumers would be willing to pay and what they actually pay for a given good or service.

The impacts of the action would depend on the terms of any Amendment 7 agreements in 2015. Under the least-constraint scenario, the expected effects would be the same as under the no-action alternative, which would not affect consumer surplus.

Under the most-constraint scenario, the action could affect consumer surplus if the bigeye tuna catch limit is reached and constrains fishing by the Hawaii longline fleet. A closure could lead to a reduced supply of local fresh bigeye tuna in the Hawaii market. This may lead to higher prices for locally caught fresh bigeye tuna or substitutes such as bigeye tuna from other sources. Although some consumers in Hawaii probably prefer locally caught bigeye tuna over bigeye tuna from other sources, if acceptable substitutes are available for the time period during which locally caught bigeye tuna is in short supply, this may dampen the impacts to consumers. If, on

the other hand, there are constraints to switching sources – for example, if the short duration of a fishery closure makes it impractical for wholesale/retail buyers to establish connections with alternative suppliers, thus limiting the availability of acceptable substitutes, then a fishery closure could result in more limited choices about product sources and higher prices, with consequent adverse impacts on consumer surplus. There is some seasonality in supply and demand of locally caught bigeye tuna. A closure at the end of the year would encompass part of the holiday season, when demand is relatively high.

Although it is difficult to predict the likelihood or magnitude of these effects on consumer surplus, it seems reasonable to conclude that the degree to which the market would shift to alternative sources is positively correlated with the length of the period during which bigeye tuna landings would be prohibited. That period is a function of the size of the limit (relative to the amount expected to be caught under no action), the magnitude of any race to fish effect that might occur as a result of the action (i.e., if imposition of the limit causes fishing to occur earlier in the year than it otherwise would, the limit would be reached that much faster), and the terms of any Amendment 7 agreements. The likelihood of shifting to alternative sources would also depend on buyers' expectations regarding closures in future years, which are not known.

The above theoretical assessment of potential impacts on consumer surplus can be supplemented with observations of how the fleet and market responded to the bigeye tuna fishery closure in 2010 (November 22 through December 31), as reported by Richmond et al. (2012; 2015). Although what occurred in 2010 is informative with respect to what would occur under the most-constraint scenario of this action, it is cautioned that none of the outcomes in 2009-2014 can be affirmatively linked as effects caused by the 2009, 2012, or 2013 rules. In other words, although we have reliable measures of much of what actually occurred in 2009-2014 (e.g., in terms of fishing effort, catches, and prices), we do not have anything other than theoretical predictions of what would have occurred under no-action, so even a post-event assessment of the effects of the action would necessarily be somewhat speculative. Furthermore, the 2010 experience documented by Richmond et al. involved a fishery closure lasting little more than a month. As described below, the proposed action could result in a closure lasting as many as five months, so the 2010 experience might have limited applicability in terms of examining the impacts of a longer closure in 2015.

Examining the expectations of the Hawaii fishing community in advance of the closure, Richmond et al. (2012; 2015) found that the biggest fear among fishermen, buyers, and retailers was that the closure would lead to a dramatic decline in the supply and a compensatory increase in price of local bigeye tuna. The study found that the impacts of the closure on the supply of locally caught bigeye tuna was not as great as anticipated by many in the fishing community. Vessels longlining for bigeye tuna in the EPO and dual-permit longline vessels increased their bigeye tuna production after the closure. Those two sources, along with vessels in the Hawaii troll and handline fisheries, provided a continuing supply of local bigeye tuna to the Hawaii market during the closure. The production of bigeye tuna, in numbers of fish, of the Hawaii longline fleet (including EPO-caught fish) during December 2010 was 23 percent less than the average production during the five previous Decembers. The average weight of the fish was 11 percent less than the average weight from the previous five Decembers (possibly linked to differing biological conditions in the EPO). Smaller bigeye tuna tend to be of lower value than

larger fish, but the average price of longline-caught bigeye tuna during December 2010 was found to be 33 percent greater in December 2010 than the average from the previous five Decembers. Richmond et al. (2012; 2015) also found that most tuna buyers reported the average quality of tuna entering the auction to have declined significantly over the course of the closure, presumably due to the longer hold times involved in trips to the EPO. The study found indications that during the closure there was an increase in the amount of lower quality bigeye tuna (and lower prices for those grades) and a decrease in availability of high- and medium-grade bigeye tuna (and higher prices for those grades). Information gathered in the study suggested that impacts to consumers in Hawaii might have been greater (i.e., prices might have been greater) if it were not for the severe snowstorms on the U.S. east coast about the same time, which hindered shipments to the U.S. east coast, leaving a greater supply in the Hawaii market.

In the event of a WCPO closure in 2015, it can be expected that, as in 2010, bigeye tuna would continue to be supplied to consumers in Hawaii and elsewhere from the longlining in the EPO, from U.S. WCPO fisheries other than longline fisheries, and from foreign sources. However, if there is a closure in 2015 it could be much longer than the 2010, and the 500 mt limit for large longline vessels could be reached, so the EPO could be a less important source than previously, and thus constrain supply and pressure prices accordingly.

Producer impacts:

Producer surplus is the difference between producers' (e.g., fishing businesses') revenues and their costs.

As described above, the action could lead to an increase in the size of the WCPO stocks of bigeye tuna and other species caught in the U.S. longline fishery, which could enhance the ability of U.S. fishermen, including those in longline fisheries, to benefit from the stocks. However, any such benefits would be minor and relatively short-lived, if they occur at all. At the same time, the action can be expected to adversely impact participants in the U.S. WCPO longline fisheries by foreclosing certain fishing opportunities. Like the potential benefits of the action, these impacts would be relatively short-lived, but they could be substantial, as described below.

Because of substantial uncertainty with respect to key factors in play in 2015, including the terms of any Amendment 7 agreements, bigeye tuna catch rates, and fishermen's responses to a fishery closure resulting from reaching the bigeye tuna catch limits, this analysis relies substantially on qualitative assessments.

In the following table and figures are most of the data used in this analysis. Table 1 shows, for the years 2005-2014, the amounts of bigeye tuna and swordfish retained in the affected U.S. longline fisheries (not including the longline fisheries of the three U.S. Participating Territories), broken down where relevant by ocean area (western and central Pacific Ocean, or WCPO, versus eastern Pacific Ocean, or EPO) and deep-set versus shallow-set.

Figure 2 and Figure 3 are illustrations of the seasonality of bigeye tuna catches in the affected U.S. longline fisheries (not including the longline fisheries of the three U.S. Participating Territories). The bigeye tuna catch limits established in the 2009, 2012 and 2013 LL rules (applicable to the U.S. longline fisheries in the WCPO) were effective in 2009-2014.

Table 1. Vessel numbers and estimates of bigeye tuna and swordfish kept in the U.S. longline fisheries, 2005-2014.

Year	Number of active vessels	Number of dual permit vessels*	Bigeye tuna retained from WCPO (deep set and shallow set) (mt)	Bigeye tuna retained from EPO (deep set and shallow set) (mt)	Bigeye tuna retained – EPO as % of total	Bigeye tuna retained from WCPO – deep set (mt)	Bigeye tuna retained from WCPO – shallow set (mt)
2005	125	NA	4,480	532	11%	4,415	65
2006	127	10	4,360	83	2%	4,306	54
2007	129	12	5,346	441	8%	5,305	40
2008	129	11	4,649	1,310	22%	4,591	58
2005-08 ave			4,709	592	11%	4,654	54
2009	127	12	3,741	729	16%	3,709	32
2010	124	13	3,576	1,356	27%	3,535	42
2011	129	20	3,559	1,050	23%	3,525	34
2012	129	20	3,634	875	19%	3,612	22
2013	136	16	3,654	2,054	36%	3,635	18
2014	141	19	3,815	2,100	36%	3,801	14

* "Dual permit vessel" means a vessel with both a Hawaii Longline Limited Access Permit and an American Samoa Longline Limited Access Permit.

Source: NMFS unpublished estimates, provided by the Pacific Islands Fisheries Science Center based on numbers of fish caught by date of capture from vessel logbook data, and average fish weights derived from landings data; weight estimates are subject to change as estimation methods are improved to better account for average weight variations by area, season, and fisheries sector.

Notes: Some of these estimates differ from previously published estimates because of subsequent updates to the data and/or methods used to make the estimates. Some data are excluded from these estimates so as not to reveal confidential data.

Figure 2. Estimates of bigeye tuna kept in the U.S. longline fisheries, WCPO only, deep-set and shallow-set, cumulative by month, 2005-2014.

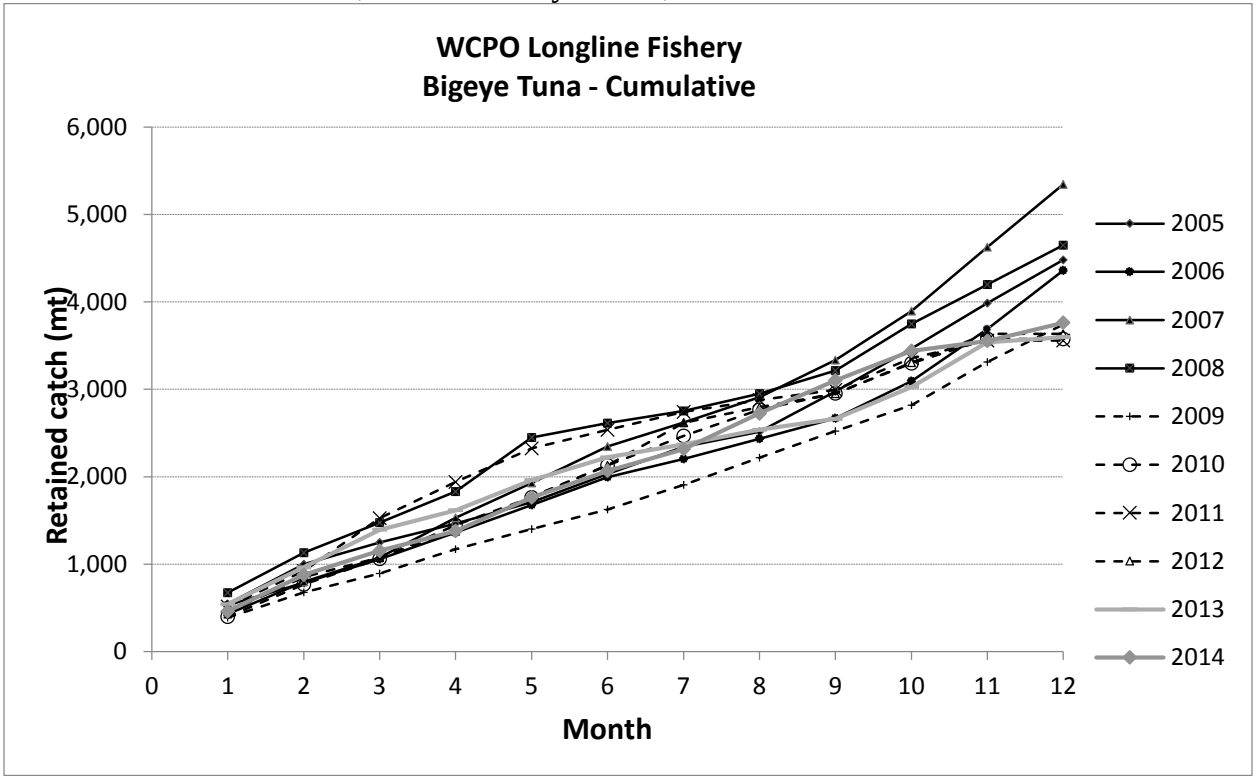
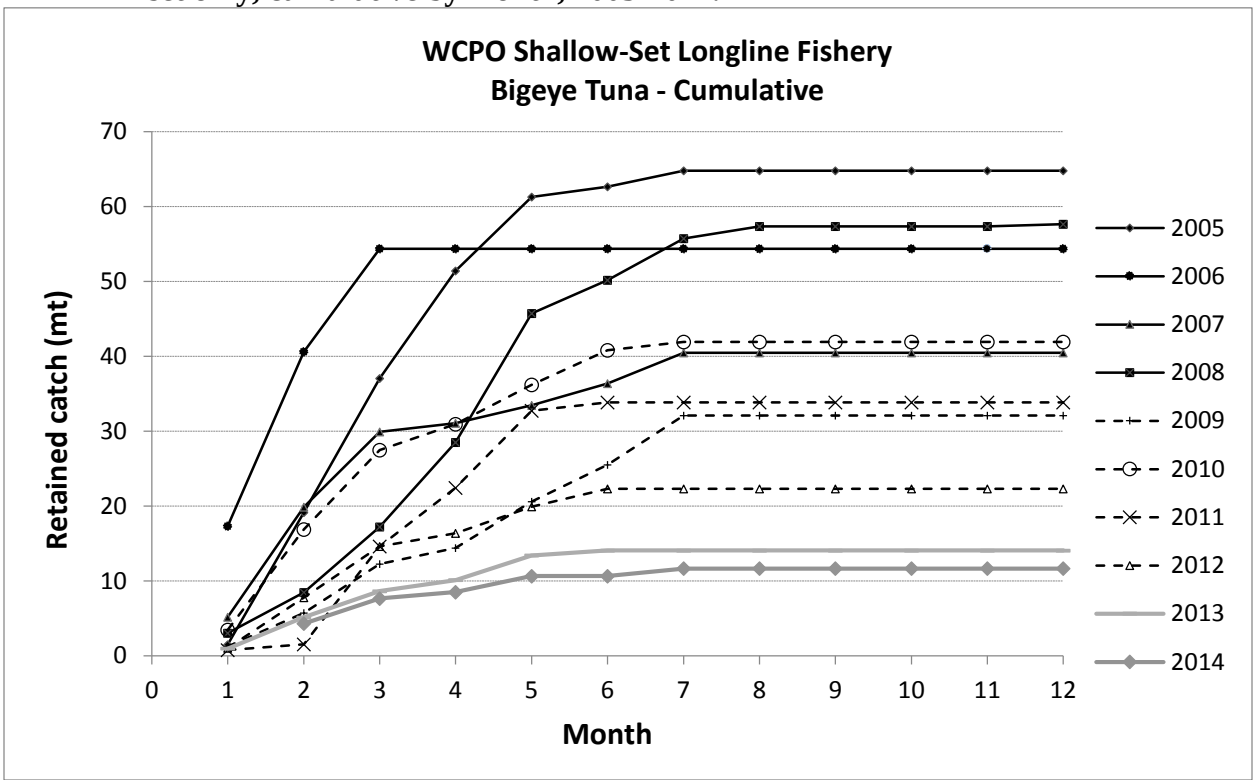


Figure 3. Estimates of bigeye tuna kept in the U.S. longline fisheries, WCPO only, shallow-set only, cumulative by month, 2005-2014.



Again, the economic effects of the 2015 bigeye tuna catch limit would depend greatly on the terms of any Amendment 7 agreements. An Amendment 7 agreement was in place in 2014 between the government of the CNMI and certain vessel owners. Section 113 arrangements were in place in 2013 (CNMI), 2012 (American Samoa) and 2011 (American Samoa). NMFS cannot predict whether or how many Amendment 7 agreements will be in place during 2015, when they go into effect, or what their terms will be, so it is difficult to assess the likely impacts of the catch limit. However, it is possible to identify the bounds of the likely range of economic effects.

At one extreme of the range, no Amendment 7 agreements would be in place in 2015, or their terms would be such that they would not affect the attribution of any bigeye tuna catches. In this case, the expected impacts of the action could be similar to those of the bigeye catch limits that were in place for 2009 and 2010, before the CFCAA or Amendment 7 was in effect. However, the limits in this action are somewhat smaller than the 3,763 mt limits that were in place in each of those two years, and as described below, bigeye catches so far in 2015 are substantially greater than in 2009, 2010, or any previous years, so the economic impacts of this action could be substantially greater.

At the other extreme, one or more Amendment 7 agreements would be in place in 2015 and their terms would be such that the U.S. longline fleet is effectively unconstrained by the catch limit established by this action. In this case, the net economic impacts would be expected to be roughly the same as under the no-action alternative (but there could be some distributional changes in net benefits, described in section 6.2).

These two extreme scenarios, which are used to identify the likely bounds of the possible impacts of the action, are referred to as the "least-constraint" scenario and the "most-constraint" scenario.

The remainder of this analysis of changes in net benefits focuses on describing the expected impacts under the least-constraint and most-constraint scenarios, reviewing what occurred in 2009-2014 when catch limits were in place, qualifying those findings as appropriate for this action, and examining, to the extent possible, where within the range of the least-constraint and most-constraint extremes the economic effects of this action are likely to lie.

Least-constraint scenario: Under this scenario, the effects would be expected to be roughly the same as the effects of the no-action alternative, which would not affect producer surplus (but some caveats are described below).

Most-constraint scenario: Under this scenario, the direct effect of the action is that it could foreclose fishing opportunities for vessels in the Hawaii longline fisheries: Owners and operators of such vessels would have to cease retaining, landing, and transshipping Convention Area-caught bigeye tuna if and when the limit is reached, for the remainder of each of the calendar years 2015. Owners and operators of vessels in the Hawaii fleet that also have an American Samoa Longline Limited Access Permit ("dual permit" vessels) would not be so constrained, but they would not be able to retain bigeye tuna caught in the portion of the U.S. EEZ surrounding

the Hawaiian Archipelago.

The economic impacts of the closure on producer surplus would depend in large part on when the limit is reached and how fishermen respond to the closure – in particular, what alternative activities they engage in during the closure. Both of these aspects are discussed below, using historical patterns as the basis for predicting likely future patterns. Both the 2005-2008 period – prior to the establishment of the bigeye tuna catch limit – and the 2009-2014 period, when bigeye tuna catch limits were in effect, are useful in this respect.

With respect to when the limit would be reached, the historical patterns are summarized in Figure 2, which shows cumulative (by month) WCPO bigeye tuna catches over the course of each of the years 2005-2014.

As reflected in Figure 2, bigeye tuna catches from 2005 through 2014 suggest a high likelihood of the 2015 limit being reached (again, this most-constraint bound assumes that any Amendment 7 agreements are such that they have no effect on the attribution of bigeye tuna among the fisheries of the United States and those of the three U.S. Participating Territories). If fishing patterns in 2015 were similar to the typical patterns in 2005-2014, 3,502 mt of bigeye tuna would be caught by some point in the last three months of each year. However, the latest available information for 2015 indicates that bigeye tuna catches so far have been substantially greater than in previous years, and that 3,502 mt are likely to be caught by early August 2015.[4]

One potentially important factor with respect to when the limits would be reached is the number of vessels with both Hawaii and American Samoa longline permits ("dual permit vessels"), because the catches of such vessels—except those in the EEZ around the Hawaiian Archipelago—are attributed to the American Samoa longline fishery, unless the vessel is in an Amendment 7 agreement with another territory. The more dual permit vessels there are, the later in the year the limit is likely to be reached. As shown in Table 1, the number of dual permit vessel has generally increased since 2009, and in 2014 stood at 19. NMFS cannot predict whether or by how much the number of dual permit vessels will change in 2015, but the establishment of the limits would create an incentive for the number to increase (indeed, the increase since 2009 might be attributable to the existence of limits starting in that year).

Another potentially important factor is that the establishment of the limits—which would be competitive limits—could cause fishermen to fish harder and/or earlier in the year than they otherwise would (i.e., if the limit results in a "race to fish"). Such an effect would result in the limit being earlier than predicted solely by pre-2009 fishing patterns. The seasonal fishing patterns in 2009-2014 do not appear to differ markedly from those prior to 2009, suggesting that this was not an important factor in those years, but as indicated above, the pattern so far in 2015 is markedly different than in the past. Whether the greater catches so far in 2015 are due in part to a race-to-fish effect is not known.

[4] Unpublished forecast prepared by the NMFS Pacific Islands Fisheries Science Center (PIFSC Internal Report IR15-22; issued July 7, 2015).

A brief summary of the consequences of the limits established for 2009-2014 follows:

- In 2009 and 2010, the annual limits of 3,763 mt were projected to be reached and the WCPO bigeye tuna fishery was closed, December 29 and November 22, respectively.

- In 2011, with a Section 113 arrangement in place with American Samoa, catches by most vessels in the Hawaii fleet started being attributed to the American Samoa longline fishery starting November 19, and the limit of 3,763 mt was not reached. Had there been no Section 113 arrangement, the limit would have been projected to be reached November 17.

- In 2012, with a Section 113 arrangement in place with American Samoa, catches by most vessels in the Hawaii fleet started being attributed to the American Samoa longline fishery starting November 20, and the limit of 3,763 mt was not reached. Had there been no Section 113 arrangement, the limit would have been projected to be reached November 27.

- In 2013, with a Section 113 arrangement in place with the CNMI, catches by most vessels in the Hawaii fleet started being attributed to the CNMI longline fishery starting December 5, and the limit of 3,763 mt was not reached. Had there been no Section 113 arrangement, the limit of 3,763 mt would have been projected to be reached in early December.

- In 2014, with an Amendment 7 agreement in place with the CNMI, catches by most vessels in the Hawaii fleet started being attributed to the CNMI longline fishery starting November 8. However, there was a limit of 1,000 mt on such transfers, which was reached, resulting in the catches at the end of the year by many of the vessels in the agreement being attributed to the U.S. fishery. The fishery was not closed, but NMFS' final estimates of 2014 catches revealed that the limit of 3,763 mt had been exceeded by 52 mt.

If the bigeye tuna limit is reached before the end of 2015 and the WCPO bigeye fishery is consequently "closed" for the remainder of the calendar year, it can be expected that affected vessels would shift to the next most profitable fishing opportunity (which might be not fishing at all). Revenues from that "next best" alternative activity reflect the opportunity costs associated with longline fishing for bigeye tuna in the Convention Area. The economic cost of the action would not be the direct losses in revenues that would result from not being able to fish for bigeye tuna in the Convention Area, but rather the difference in benefits derived from that activity and those derived from the next best activity. The economic cost of the action on affected entities is examined here by first estimating the nominal direct losses in revenues that would result from not being able to fish for bigeye tuna in the Convention Area as a result of the catch limit being reached. Those losses represent the upper bound of the economic cost of the action for producers. Potential next-best alternative activities are then identified in order to provide a (mostly qualitative) description of the degree to which actual costs would be lower than that upper bound.

Upper bounds on potential economic costs can be estimated by examining the projected value of longline landings from the Convention Area that would not be made as a result of reaching the limit. Given that the rate of bigeye tuna catches so far in 2015 is unprecedented, it is difficult to predict what total 2015 catches would be in the absence of a limit. Based on the latest available information, NMFS projects that the proposed limit of 3,502 mt will be reached on or about August 7, 2015.[4] If the bigeye tuna catch patterns after that date follow the average patterns during 2007-2014, NMFS estimates that about 5,435 mt would be caught by the end of 2015.[4] Thus, under those assumptions, imposition of the proposed limit of 3,502 mt would result in a total 2015 catch about 1,933 mt less, or 36 percent less, than the total catch expected in the absence of a limit.

In the Hawaii deep-set fishery, catches of marketable species other than bigeye tuna would likely be affected in a similar way (if vessels do not shift to alternative activities). Assuming that ex-vessel prices would not be affected by a fishery closure, revenues in the deep-set fishery 2015 could be—given the assumptions given above—36 percent less than if no limit were in place. Again, because the rate of bigeye tuna catches so far in 2015 is unprecedented, and lacking data on catches so far in 2015 for species other than bigeye tuna and on ex-vessel prices for bigeye tuna and other species, it is difficult to estimate what ex-vessel revenues would be in 2015 in the absence of a limit. If average annual ex-vessel revenues during 2005-2008, about $66.5 million (in 2014 dollars; derived from WPRFMC 2014) are a good indicator of 2015 revenues under no-action, average per-vessel revenues during 2015 under the proposed limit could be $23.9 million (36 percent of $66.5 million) less than if there were no limit. But the extraordinary bigeye catch rate so far in 2015 suggests that total 2015 revenues in the fishery in the absence of a limit could be substantially greater than revenues in 2005-2008, in which case the impacts of the limit would be greater, and if bigeye tuna catches were to continue at an extraordinarily high rate through the remainder of 2015 in the absence of a limit, the impacts could be greater yet.

In the shallow-set fishery, economic impacts would be limited if the limit is reached. The cost would be approximately equal to the revenues lost from not being able to retain or land bigeye tuna captured while shallow-setting in the Convention Area, or the cost of shifting to shallow-setting in the EPO, whichever is less. Almost all shallow-setting effort in the last quarter of the year typically takes place in the EPO. For example, in the fourth calendar quarters of 2005-2008 (before any bigeye tuna catch limits were in place in the Convention Area), almost all shallow-setting effort took place in the EPO, and 97 percent of bigeye tuna catches were made in the EPO. In other words, the catch of bigeye tuna in the shallow-set longline fishery in the Convention Area is highly seasonal, and relatively little is caught there in the latter part of the year. Thus, the cost to the shallow-set fishery of a bigeye tuna fishery closure would appear to be small. During 2005-2008, the shallow-set fishery caught an average of 54 mt of bigeye tuna per year from the Convention Area (Table 1). If the bigeye tuna catch limit is reached even as early as July 31, 2015, the Convention Area shallow-set fishery would have caught at that point, based on 2005-2008 data, on average, 99 percent of its average annual bigeye tuna catches (see Figure 3). Thus, imposition of the restrictions at that point in 2015 would result in the loss of revenues from approximately 0.5 mt (1 percent of 54 mt) of bigeye tuna, which, based on recent ex-vessel prices, would be worth no more than $5,000. The remainder of this analysis focuses on potential impacts on producer surplus in the deep-set fishery.

With respect to the mitigating effect of most catches of dual permit vessels not being subject to the limit, the RIR for the 2009 LL rule (NMFS 2009) found that the then-proposed action would establish incentives that would likely both tend to increase the number of dual permit vessels and affect their fishing activity. Indeed, the number of such vessels increased from 11 in 2009 to 20 in 2011, and it was 19 in 2014 (Table 1), and Richmond et al. (2012; 2015) found that the bigeye tuna landings of dual permit vessels increased substantially after the start of the closure in 2010. Thus, the economic losses in the most-constraint scenario would likely be somewhat less than described above, which accounted for the revenue of dual permit vessels that land their catch in Hawaii.

With respect to the possibility of price effects either mitigating or exacerbating losses in revenues, the traditional seasonality in supply and demand of locally caught bigeye tuna is relevant. A closure at the end of the year would encompass part of the holiday season, when prices are relatively high. In 2010, the limit was projected to be reached and the fishery closed on November 22. Richmond et al. (2012; 2015) found the ex-vessel total weight of longline-caught bigeye tuna sold in December 2010 (including from the EPO) to be 32 percent less than the average of the previous five Decembers (both the number of fish and average weight per fish were lower than the historical average), but the average price was 50 percent greater, so total ex-vessel value was only 4 percent less than the historical total December value.

As stated previously, actual economic losses are likely to be less than the upper bounds described above because ceasing fishing would not necessarily be the most profitable opportunity in the event of the catch limit being reached. Two alternative opportunities that are expected to be attractive to affected entities include: (1) deep-set longline fishing for bigeye tuna in the Convention Area in a manner such that the vessel is considered part of the longline fishery of American Samoa, Guam, or the CNMI; and (2) deep-set longline fishing for bigeye tuna and other species in the EPO. These two opportunities are discussed in detail below. Three additional opportunities, which were examined in economic analyses prepared for the 2009 and 2012 rules are: (3) shallow-set longline fishing for swordfish (for deep-setting vessels that would not otherwise do so), (4) deep-set longline fishing in the Convention Area for species other than bigeye tuna, and (5) working in cooperation with vessels operating as part of the longline fisheries of the Participating Territories – specifically, receiving transshipments at sea from them and delivering the fish to the Hawaii market. Vessel repair and maintenance is another possibility. A study of the 2010 WCPO bigeye tuna fishery closure (Richmond et al. 2012; 2015) did not identify any alternative activities that vessels engaged in during the closure other than deep- setting in the EPO, vessel maintenance and repairs, and granting lengthy vacations to employees. Thus, alternative opportunities (3), (4) and (5) are probably relatively unattractive relative to the first two, and they are not discussed here in any further detail.

Before examining in detail the two potential alternative opportunities that would appear to be the most attractive to affected entities, it is important to note that under the action, once the limit is reached and the WCPO bigeye tuna fishery is closed, it would be prohibited to fish with longline gear both inside and outside the Convention Area during the same trip (with the exception of a fishing trip that is in progress when the limit is reached and the restrictions go into effect). For example, after the restrictions go into effect, during a given fishing trip, a vessel could be used for longline fishing for bigeye tuna in the EPO or for longline fishing for species other than

bigeye tuna in the Convention Area, but not both. This reduced operational flexibility would bring costs, since it would constrain the potential profits from alternative opportunities collectively. Those costs cannot be quantified.

With respect to alternative opportunity (1), deep-setting for bigeye tuna in a manner such that the vessel is considered part of the longline fishery of one of the three U.S. Participating Territories, there would be three such ways to do so: (a) landing the bigeye tuna in one of the three Participating Territories; (b) having an American Samoa Longline Limited Access Permit; or (c) entering into an Amendment 7 agreement with one or more of the three Participating Territories, such that the vessel is considered part of the Participating Territory's longline fishery. In the first two circumstances, the vessel would be considered part of the longline fishery of the Participating Territory only if the bigeye tuna were not caught in the portion of the U.S. EEZ around the Hawaiian Islands and they are landed by a U.S. vessel operated in compliance with a permit issued under the Pelagics FEP or the West Coast HMS FMP.

With respect to alternative opportunity (1)(a), landing the bigeye tuna in one of the Participating Territories, there are three potentially important constraints. First, whether the fish are landed by the vessel that caught the fish or by a vessel to which the fish were transshipped, the costs of a vessel transiting from the traditional fishing grounds in the vicinity of the Hawaiian Archipelago to one of the Participating Territories would be substantial. Second, none of these three locales has large local consumer markets to absorb substantial additional landings of fresh sashimi-grade bigeye tuna. Third, transporting the bigeye tuna from these locales to larger markets, such as in Hawaii, the U.S. west coast or Japan, would bring substantial additional costs and risks. These cost constraints suggest that this opportunity has limited potential to mitigate the economic impacts of the rule on affected small entities.

Opportunity (1)(b), having an American Samoa Longline Limited Access Permit, would be available only to the subset of the Hawaii longline fleet that has both Hawaii and American Samoa longline permits ("dual permit vessels"). Vessels that do not currently have both permits could obtain them if the permits are available, they meet the eligibility requirements and they pay the required costs. For example, the number of dual permit vessels increased from 12 in 2009, when the first WCPO bigeye tuna catch limit was established, to 20 in 2011, and it was 19 in 2014 (Table 1). Richmond et al. (2012; 2015) found that bigeye tuna landings of dual permit vessels increased substantially after the start of the closure on November 22, 2010, indicating that this was an attractive opportunity for dual permit vessels, and suggesting that those businesses might have benefitted from the catch limit and the closure.

Opportunity (1)(c), entering into an Amendment 7 agreement, was also available in 2011-2014 (in 2011-2013, actually under Section 113 of the CFCAA). As a result of arrangements that were in place in 2011-2014, the bigeye tuna fishery was not "closed" in any of those four years. Participation in an Amendment 7 agreement would likely not come without costs to fishing businesses. As an indication of the possible cost, the terms of the arrangement between American Samoa and the members of the Hawaii Longline Association that applied in 2011 and 2012 included payments totaling $250,000 from the Hawaii Longline Association to the Western Pacific Sustainable Fisheries Fund. However, any payments made by participating businesses under the terms of Amendment 7 agreements would likely be made within the U.S. economy, in

which case there would be no change in net benefits to the nation (but see section 6.2 for distributional changes in net benefits). Again, if enough vessels enter into Amendment 7 agreements, such that the catch limit ceases to be constraining on the fleet, this would result in the least-constraint scenario described above. However, Amendment 7 stipulates that the agreements be funded by deposits to the Western Pacific Sustainable Fisheries Fund in support of fisheries development projects identified in a territory's Marine Conservation Plan adopted pursuant to Section 204 of the MSA. This constraint on how the funds can be used can be expected to diminish their value. Thus, even under the least-constraint scenario, the catch limits could result in adverse economic impacts.

With respect to alternative opportunity (2), deep-set fishing for bigeye tuna in the EPO, this would be an option for fishing businesses only if it is allowed under regulations implementing the decisions of the Inter-American Tropical Tuna Commission (IATTC). Currently there is a bigeye tuna catch limit of 500 mt for 2015 and 2016 that applies to U.S. longline vessels greater than 24 meters (m) in length (50 CFR 300.25). It cannot be predicted with any certainty whether the limit will be reached in 2015. Annual longline bigeye tuna catch limits have been in place for the EPO in most years since 2004. Since 2009, when the limit was 500 mt and applicable only to vessels longer than 24 m in length, the limits were reached in 2013 (November 11) and 2014 (October 31). The latest available data for 2015 indicate that similar to the situation in the Convention Area, bigeye tuna catches in the EPO so far in 2015 are at an unusually high level—cumulative catches through mid-June by longline vessels at least 24 m in length are about twice as great as the typical level by that date in the previous four years.[4] The highly seasonal nature of bigeye tuna catches in the EPO and the relatively high inter-annual variation in catches prevents NMFS from making a useful prediction of whether and when the limit is likely to be reached, but there appears to be a substantial likelihood that in 2015, retaining bigeye tuna in the EPO will be prohibited for large longline vessels for a substantial portion of the latter part of the year. This would constrain the mitigating potential for this alternative opportunity for large longline vessels, which constitute about a quarter of the fleet.

Historical fishing patterns can provide an indication of the likelihood of affected entities making use of the opportunity of deep-setting in the EPO in the event of a closure in the WCPO. The proportion of the U.S. fisheries' annual bigeye tuna catches that were captured in the EPO from 2005 through 2008 ranged from 2 percent to 22 percent, and averaged 11 percent (Table 1). In 2005-2007, that proportion, which ranged from 2 percent to 11 percent, may have been constrained by the bigeye tuna catch limits established by NMFS to implement the decisions of the IATTC. Prior to 2009, most of the U.S. annual bigeye tuna catch by longline vessels in the EPO typically was been made in the second and third quarters of the year: in 2005-2008 the percentages caught in the first, second, third, and fourth quarters were about 14, 33, 50, and 3 percent, respectively. These two historical patterns – that relatively little of the bigeye tuna catch in the longline fisheries was typically taken in the EPO (11 percent in 2005-2008, on average) and that most EPO bigeye tuna catches were made in the second and third quarters, with relatively few catches in the fourth quarter, suggest that there could be substantial costs for at least some affected entities to shift to deep-set fishing in the EPO in the event of a closure in the WCPO. On the other hand, fishing patterns in 2009-2014, when annual bigeye tuna catch limits were in effect in the WCPO, suggest that a substantial shift in deep-set fishing effort to the EPO could occur, and the unusually high rate of bigeye tuna catches in the EPO so far in 2015 support

the notion that fishing in the EPO has become more attractive in recent years. In 2009, 2010, 2011, 2012, 2013, and 2014, the proportions of the fisheries' annual bigeye tuna catches that were captured in the EPO were about 16, 27, 23, 19, 36, and 36 percent, respectively (Table 1). Furthermore, most bigeye tuna catches in the EPO have typically been made in the latter half of the calendar year.

The authors of the study of the WCPO bigeye tuna longline fishery closure in 2010 (Richmond et al. 2012; 2015) expressed the choice faced by Hawaii longline fishermen during the closure as being either fishing in the EPO or quitting fishing until January 1, 2011. The study found that the number of trips taken in December 2010 was 35 percent less than the average number in the previous five Decembers. Although this is a substantial reduction, it reflects a sizeable amount of fishing in the EPO, as well as shallow-setting and deep-setting in the Convention Area by dual permit vessels. However, the study found that some businesses – particularly those with smaller vessels (the implications in terms of distributional impacts are discussed in section 6.2) – were less inclined than others to fish in the EPO during the closure because of the relatively long distances that would need to be travelled in the relatively rough winter ocean conditions. The study identified a number of factors that likely made fishing in the EPO less lucrative than fishing in the WCPO during that part of the year, including fuel costs and the need to limit trip length in order to maintain fish quality and because of limited fuel storage capacity. On the other hand, as noted above, prices during the closure in December were 50 percent greater than the average during the previous five Decembers. And again, in 2013 and 2014 there was a marked increase in fishing activity in the EPO, with 36 percent of the fleet's bigeye tuna catches made there in both years.

In addition to affecting the volume of landings of bigeye tuna and other species, the catch limit could affect fish prices, particularly during a fishery closure. Both increases and decreases appear possible. After the limit is reached and landings from the WCPO are prohibited, ex-vessel prices of bigeye tuna (e.g., that are caught in the EPO or by vessels in the longline fisheries of the three U.S. Participating Territories), as well as of other species landed by the fleet, could increase as a result of the constricted supply. This would mitigate economic losses for vessels that are able to continue fishing and landing bigeye tuna during the closure. For example, Richmond et al. (2012; 2015) found that ex-vessel prices during the closure in December were 50 percent greater than the average during the previous five Decembers (it is emphasized that because it was an observational study, neither this nor other observations of what occurred during the closure can be affirmatively linked as effects of the fishery closure).

Conversely, a WCPO bigeye tuna fishery closure could cause a decrease in ex-vessel prices of bigeye tuna and other products landed by affected entities if the interruption in the local supply prompts the Hawaii market to shift to alternative (e.g., imported) sources of bigeye tuna. Such a shift could be temporary – that is, limited to 2015, or it could lead to a more permanent change in the market (e.g., as a result of wholesale and retail buyers wanting to mitigate the uncertainty in the continuity of supply from the Hawaii longline fisheries). In the latter case, if locally caught bigeye tuna fetches lower prices because of stiffer competition with imported bigeye tuna, then ex-vessel prices of local product could be depressed indefinitely. Richmond et al. (2012; 2015) found that a common concern in the Hawaii fishing community prior to the closure in November 2010 was retailers having to rely more heavily on imported tuna, causing imports to gain a

greater market share in local markets. The study found this not to have been borne out, at least not in 2010, when the evidence gathered in the study suggested that few buyers adapted to the closure by increasing their reliance on imports, and no reports or indications were found of a dramatic increase in the use of imported bigeye tuna during the closure. The study concluded, however, that the 2010 closure caused buyers to give increased consideration to imports as part of their business model, and it was predicted that tuna imports could increase during any future closure. As noted above, the likelihood of this occurring can be expected to be positively correlated with the length of a closure and the degree to which the closure constrains the supply from the local fleet. The likelihood would also depend on buyers' expectations regarding closures in future years, which are not known. To the extent that ex-vessel prices would be reduced by this action, revenues earned by affected entities would be affected accordingly, and these impacts could occur both before and after the limit is reached, and as described above, possibly after 2015.

Public sector costs:

Implementation of the action would result in federal government costs.

The U.S. government—particularly NMFS and the U.S. Coast Guard—would bear the responsibility and costs of enforcing the prohibitions that would go into effect if and when the limit is reached. The costs of the on-the-water and on-the-ground aspects of such enforcement would probably be minimal, as they would be largely conducted in the course of routine patrols and surveillance activities used to enforce a variety of laws. There would also be costs associated with outreach to affected vessel owners and operators.

NMFS would need to monitor bigeye tuna catches with respect to the limits, which would involve faster and more intensive data processing and analysis that would otherwise be conducted in the absence of limits. NMFS would not necessarily have to do this year-round; it could focus on those portions of each year when there is an appropriately high likelihood of the limit being reached within a certain period. Using the monitoring results, NMFS would have to make determinations as to whether and when the limit is likely to be reached, and if such a determination is made, prepare and publish a notice in the *Federal Register* that announces the effective date of the bigeye tuna retention, landing, and transshipment restrictions.

Summary of effects on net benefits:

As described above, the action can be expected to have neutral or minor positive incremental effects (relative to no action) on benefits that the United States can potentially enjoy through the maintenance of productive WCPO bigeye tuna and other species caught in the longline fishery. Those effects, however, are likely to be minor and relatively short-lived. Such effects would be more sizeable if other fishing nations in the WCPO implement similar actions and they would be more lasting if the United States and other fishing nations in the WCPO implement similar conservation actions beyond the one-year duration of this action (see discussion of cumulative effects in section 6.4).

Any positive incremental effects of the action would be offset by incremental costs to the nation in terms of lower producer surplus and possibly lower consumer surplus that would occur if the longline fishery is closed, and incremental public sector costs. These impacts cannot be quantifiably predicted with any certainty, and would depend greatly on whether there are any Amendment 7 agreements and their terms. An upper-bound analysis indicates that gross revenues in the deep-set longline fishery in 2015 could be at least $24 million less under the proposed limit than in the absence of a limit. These direct impacts would occur only in 2015, although there is a possibility of more lasting costs in the longline sector if the bigeye tuna catch limit in 2015 contributes to more lasting shifts in Hawaii's seafood market. It is foreseeable that similar actions would be taken in the future years, in which case the cumulative costs could be substantial and lasting (see discussion of cumulative effects in section 6.4).

It is not possible to determine whether the benefits of the action would outweigh the costs.

6.2. Distributional changes in net benefits

Examples of distributional effects include differential economic impacts according to geographical region and businesses of differing sizes.

The final rule would apply only to the U.S. WCPO longline fishery (i.e., essentially the Hawaii longline fishery). Fisheries involving other vessel types and in other areas of the United States would be unaffected. To the extent that the action results in the stocks of WCPO bigeye tuna and other species larger than they otherwise would be (any effects of which would be minor and short-lived), other U.S. fisheries in the Pacific Ocean that exploit the stocks would benefit without bearing any costs.

As indicated in 6.1, it is possible that as a result of a post-limit fishery closure, the Hawaii market could switch to alternative sources of bigeye tuna. Such alternative sources could be both foreign and domestic. For example, as described above, fishing businesses in the Hawaii troll and handline fisheries appeared to have responded to the 2010 bigeye tuna longline fishery closure by increasing their production of bigeye tuna during the closure. Thus, the action – if it results in a fishery closure – could result in a shift in benefits to businesses in fisheries like the Hawaii troll and handline fleets. If the alternative sources of bigeye tuna are foreign, there would be no distributional changes (within the United States) in net benefits.

It is possible that the action would lead to a shift of deep-set longline fishing effort into the EPO. However, the operation of the vessels involved in such shifts is not expected to change in terms of where the vessels are based or where they land their catch, so no distributional economic effects are expected as a result of such shifts.

As described in section 6.1, the action would have different impacts on dual permit vessels relative to other vessels in the Hawaii longline fleet. Richmond et al. (2012; 2015) found that the rate at which bigeye tuna were caught and kept by dual permit vessels increased after the start of the 2010 closure, indicating that those fishing businesses may have benefited from the closure and the action in general.

As described in section 6.1, the action could have different impacts on vessels that engage in the shallow-set fishery relative to those that participate in the deep-set fishery. While virtually all affected vessels engage at least part of the year in the deep-set fishery, less than half of the Hawaii longline fleet has engaged in the shallow-set fishery since 2005 (NMFS unpublished data).

The action would not have any direct distributional impacts directly in terms of the sizes of affected businesses. However, it could have indirect distributional impacts to vessels of different sizes, which might be correlated with business size. First, the 500 mt EPO bigeye catch limit for 2015 applies only to vessels greater than 24 m in length, so in the event that the WCPO bigeye tuna fishery is closed and the 500 mt limit is reached in the EPO, only vessels 24 m or less in length would be able to take advantage of the alternative opportunity of deep-setting for bigeye tuna in the EPO. On the other hand, Richmond et al. (2012; 2015) found that during the 2010 WCPO bigeye tuna fishery closure some businesses – particularly those with smaller vessels – were less inclined than others to fish in the EPO during the closure because of the relatively long distances that would need to be travelled in the relatively rough winter ocean conditions. Thus, smaller vessels might be expected to find it more difficult, risky, and/or costly to fish in the EPO during the relatively rough winter months than larger vessels.

The action would have different impacts on businesses whose vessels are included in Amendment 7 agreements relative to businesses whose vessels are not. The fishing activities of vessels in the former category would be less constrained than vessels in the latter category, and in some circumstances entirely unconstrained by the bigeye tuna catch limit, and the economic effects on the two categories of businesses would differ accordingly. On the other hand, as described in section 6.1, participation in an Amendment 7 agreement is expected to bring costs to fishing businesses that participate in it (NMFS does not know who would bear the payments under Amendment 7 agreements, but presumes that fishing businesses participating the agreements would bear costs). Whether the benefits of participation would outweigh the costs cannot be predicted. In either case, these differential impacts within the Hawaii longline fleet (between those businesses that participate in Amendment 7 agreements and those that do not) should not be attributed to the action because the action would not compel or prohibit any fishing businesses to participate in Amendment 7 agreements.

A potential distributional effect of this action in combination with Amendment 7 is that longline fishing sector participants might, under Amendment 7 agreements, make payments to the Western Pacific Sustainable Fisheries Fund in support of fisheries development projects identified in a territory's Marine Conservation Plan (MCP). The territories would consequently stand to benefit from the implementation of their MCPs, and the fishery participants would potentially receive compensation through relief from the post-limit fishing restrictions. It is not possible to quantitatively predict these net distributional effects.

6.3. Changes in income and employment

It is not possible to quantify impacts in terms of income and employment.

To the extent that the action would adversely impact U.S. WCPO longline fishing businesses,

those impacts could be passed on to vessel crew members in the form of reduced compensation, and/or to business sectors with backward linkages to the producers, such as businesses that supply the fishing vessels. This could also be true for business sectors with forward linkages to the producers, such as businesses in Hawaii that market the longline fleet's catches. Such impacts would be expected to be limited to forward-linked businesses that do not fill the gap in local product with bigeye tuna sourced elsewhere (see section 6.1). Furthermore, if such substitution does occur, it might not occur at the point of ex-vessel sales. For example, the businesses that buy from the longline vessels (the United Fishing Agency's fish auction in Honolulu being the primary one at present) might not seek alternative supplies while the subsequent buyers do seek alternative supplies. In that case, the action would result in impacts to only that first level of forward-linked businesses. Richmond et al. (2012; 2015) found that many stakeholders in the longline sector believed that the effects of the impending bigeye tuna fishery closure in 2010 would ripple through the Hawaii seafood economy and possibly result in job losses in support industries, such as at the fish auction, freight companies that ship locally caught seafood abroad, and fishing supply companies. The study did not find these beliefs to be borne out, but it concluded that many bigeye tuna dealers struggled during the closure.

Using an economic input-output table developed for Hawaii and cost-earnings data for Hawaii's fisheries, Arita et al. (2013) analyzed how Hawaii's fishing sector, including the longline fishing specifically, impacts household incomes across different socioeconomic categories. The longline fishery sectors were found to have household income multipliers that are relatively small compared those of the Hawaii's non-fishery sectors. For both the deep-set and shallow-set longline sectors, a gain in household income of $0.71 is expected for every additional $1 increase in demand. With respect to socioeconomic categories, the study found that the longline sectors have a relatively weak impact on the high-income group, a relatively high impact on the middle-income group, and an impact on low-income group approximately in line with non-fishery averages.

The funding of Amendment 7 agreements could positively affect income and employment in the recipient territorial economies. The effects would depend on the funding amounts and the specific fisheries development projects that they are used to support. However, there could be counteracting adverse effects on income and employment in the payers' economies, as they would be transferring to the territories funds that could otherwise be used to generate income and employment in their own economies.

6.4. Cumulative effects

Cumulative effects are the additive effects of this action and other existing and reasonably foreseeable actions (e.g., other fishery regulations). The cumulative effects of the action can be described only qualitatively.

Potentially relevant other present (recently completed and in-progress) actions include specific actions being taken to manage U.S. HMS fisheries in the Convention Area. They include:

- U.S. implementation of the purse seine and longline provisions of the Commission's predecessor CMMs to CMM 2014-01 (see section 4); the final rule to implement

restrictions on the use of FADs for 2015 (79 FR 77942; December 29, 2014); the interim rule to establish a limit on purse seine fishing effort in the U.S. EEZ and on the high seas in the Convention Area for 2015 (80 FR 29220; May 21, 2015); and the anticipated proposed rule and proposed specifications that would establish, among other things, limits on bigeye tuna catches in the U.S. longline fishery restrictions on the use of fish aggregating devices by purse seine vessels in 2015-2017.

- U.S. implementation of the IATTC decisions for tropical tunas in the EPO in 2009, 2011, and 2013, which include bigeye tuna catch limits for longline fisheries and closed areas and periods for purse seine fishing for the years 2009 through 2016 (see final rule published November 23, 2009, at 74 FR 61046; final rule published November 4, 2011, at 76 FR 68332; and final rule published April 9, 2014, at 79 FR 19487).

- NMFS issued a final rule to implement for U.S. fishing vessels IATTC Resolution C-11-10, "Resolution on the Conservation of Oceanic Whitetip Sharks Caught in Association with Fisheries in the Antigua Convention Area" (76 FR 68332; November 4, 2011).

- NMFS issued a final rule to implement the provisions of IATTC Resolution C-11-01, on tropical tunas for 2012 and 2013, for U.S. purse seine vessels and longline vessels (76 FR 68332; November 4, 2011). The longline element is a bigeye tuna catch limit of 500 mt for each of 2012 and 2013, applicable to U.S. longline vessels more than 24 m in length. The purse seine elements include the following: a period of time when all purse seine fishing is prohibited – vessel owners can choose between two closure periods; an area of the high seas that is closed to fishing during certain times of the year; and catch retention requirements.

- NMFS issued a final rule to implement for U.S. fishing vessels IATTC Resolution C-11-10, "Resolution on the Conservation of Oceanic Whitetip Sharks Caught in Association with Fisheries in the Antigua Convention Area" (76 FR 68332; November 4, 2011). Under this rule, oceanic whitetip shark may not be retained by U.S. HMS fishing vessels in the EPO.

- NMFS issued a final rule to implement for U.S. fishing vessels IATTC Resolution C-11-03, "Resolution Prohibiting Fishing on Data Buoys" (76 FR 68332; November 4, 2011).

- Based on a Western Pacific Fishery Management Council (WPFMC) recommendation, NMFS issued a final rule on June 11, 2012 (77 FR 34260), that modifies the boundaries of the American Samoa large vessel prohibited area to align with the boundaries of the Rose Atoll Marine National Monument, effective July 11, 2012.

- NMFS issued a final rule to implement provisions of several WCPFC CMMs on December 3, 2012 (77 FR 71501). The final rule, effective January 2, 2013, establishes notice, reporting, and observer coverage requirements for transshipments, requirements regarding notification of entry into or exit from a particular area of the high seas, and requirements regarding discards from purse seine vessels.

- NMFS issued a final rule that prohibits commercial fishing in the Pacific Remote Islands and Rose Atoll Monuments, and in the Islands Units of the Marianas Trench Monument; establishes management measure for non-commercial and recreational charter fishing in the Monuments; and prohibits the conduct of commercial fishing outside the Monuments and non-commercial fishing inside the Monuments during the same trip (78 FR 32996; June 3, 2013).

- NMFS issued a final rule to implement the provisions of IATTC Resolution C-13-01, on tropical tunas, for U.S. purse seine vessels and longline vessels (79 FR 19487; April 9, 2014). The rule effectively extends through 2016 the regulations issued to implement Resolution C-11-01 (see above). The longline element is a bigeye tuna annual catch limit of 500 mt, applicable to U.S. longline vessels more than 24 m in length. The purse seine elements include the following: a seasonal period when all purse seine fishing is prohibited – vessel owners can choose between two closure periods; and an area of the high seas that is closed to fishing during certain times of the year.

- NMFS issued a final rule to implement WCPFC decisions on the oceanic whitetip shark, the whale shark, and the silky shark (80 FR 8807; February 19, 2015).

- NMFS issued a final rule to prohibit commercial fishing, while allowing for managed non-commercial fishing, in the expanded areas of the Pacific Remote Islands Marine National Monument, which includes the waters of the U.S. EEZ around Jarvis and Wake Islands and Johnston Atoll, consistent with Presidential Proclamation 9173, issued in September 2014 (80 FR 15693; March 25, 2015).

- NMFS issued a final rule to implement Amendment 7 to the Pelagics FEP that set up a system for the assignment of WCPFC-imposed HMS catch limits among the United States and American Samoa, Guam, and the Commonwealth of the Northern Mariana Islands. This action also established a catch limit of 2,000 mt of longline-caught bigeye tuna for each territory for 2014, 1,000 mt of which could be allocated to eligible U.S. longline fishing vessels (79 FR 64097; October 28, 2014).

- The parties to the SPTT agreed on an interim arrangement for 2015 that provides for access by U.S. purse seine vessels to the waters of the Pacific Island parties to the SPTT; although the total number of fishing days under the arrangement is similar to previous agreements, the number of fishing days allowed in the Kiribati EEZ is only 300.

- In 2006 Kiribati formed the Phoenix Islands Protected Area (PIPA) in its EEZ, which is about 140,000 square miles in size. On January 1, 2015, Kiribati banned all commercial fishing within a significant portion of the PIPA. This prohibition applies to the U.S. purse seine fleet.

Potentially relevant other present (recently completed and in-progress) actions include:

- Actions by other nations to implement CMM 2014-01, details of which are unknown at this time.

Reasonably foreseeable future actions include future fishery management actions by the United States and other nations:

- Actions by the United States and other nations to implement any additional management measures adopted by the Commission for resources in the affected environment, details of which are unknown at this time.

- Actions by the United States and other nations to implement a new multi-year IATTC management measure for tropical tunas for 2017 and beyond, details of which are unknown.

- Actions by the United States to implement a renegotiated SPTT, the specific details of which are unknown at this time.

Cumulative benefits:

As described in section 6.1, the action has the potential to result in the WCPO stocks of bigeye tuna and other species being slightly larger than they would be without a catch limit, but the effect would be relatively short-lived. With some exceptions, the other present and reasonably foreseeable actions, as described above, have the potential to collectively increase the sizes of these stocks. In particular, implementation by other WCPFC members of the various provisions of CMM 2014-01 would be expected to have much more substantial impacts on bigeye tuna and the other tropical tuna stocks than the action alone, since foreign fleets account for the majority of fishing mortality on the these stocks in WCPO HMS fisheries and the CMM includes provisions applicable through 2017. The cumulative benefits derived from larger stock sizes, as described in section 6.1, would be accordingly greater than those of this action alone.

Cumulative costs:

One key factor related to cumulative costs in the longline production sector is the degree to which the Hawaii market shifts to alternative sources of bigeye tuna as a result of any fishery closures like the ones that could result from this action. As noted in section 6.1, the likelihood and magnitude of such a shift can be expected to depend not just on the length of a closure in a given year, but also on seafood buyers' expectations of future fishery closures. If annual longline bigeye catch limits become routine, and if they are formulated such that they substantially limit the supply of bigeye tuna from the Hawaii longline fleet, the Hawaii market could shift to alternative (e.g., foreign) sources of bigeye tuna, with consequent cumulative impacts on consumer and producer surplus. On the other hand, the likelihood and magnitude of such a shift would be lower if fishing businesses in the Hawaii longline fleet continue to have the opportunity to operate as part of the longline fisheries of the U.S. Participating Territories through participation in Amendment 7 agreements.

Net cumulative effects:

It is not possible to gauge whether the cumulative beneficial economic effects of this action and other present and future actions would outweigh their cumulative costs.

7. DETERMINATION OF SIGNIFICANCE UNDER EXECUTIVE ORDER 12866

In accordance with E.O. 12866, NMFS has made the following determinations:

- This rule is not likely to have an annual effect on the economy of more than $100 million or to adversely affect in a material way the economy, a sector of the economy, productivity, jobs, the environment, public health or safety, or state, local, or tribal governments or communities.
- This rule is not likely to create any serious inconsistencies or otherwise interfere with any action taken or planned by another agency.
- This rule is not likely to materially alter the budgetary impact of entitlements, grants, user fees or loan programs or the rights or obligations of recipients thereof.
- This rule is not likely to raise novel or policy issues arising out of legal mandates, the President's priorities, or the principles set forth in E.O. 12866.

Based on these determinations, the rule considered in this RIR is not a "significant regulatory action" for the purposes of E.O. 12866. Furthermore, the rule is not controversial.

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